

Hyperbolic RQ-VAE enhanced Generative Recommendation with Differential-Length Codebook: An Independent Reproduction on Amazon Musical_Instruments

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Abstract

We present an independent reproduction of HG-Rec (Zhang et al., ICML 2026) on Amazon Musical_Instruments, a dataset not featured in the original paper’s headline experiments. We replicate the full pipeline Stage 1 sentence-T5 embedding, Stage 2 RQ-VAE tokenization, Stage 3 T5-small generation, Stage 4 beam-search inference and verify that the relative ranking HG-Rec > Letter > TIGER > SASRec is preserved. However, our absolute numbers are systematically 1861% lower than the paper-reported baselines across 8 methods, indicating a dataset/protocol gap that we document. Beyond reproduction, we contribute four findings: (i) per-layer curvature grid ablation shows that the optimal curvature is at or near zero (5.3% R@10 span); (ii) free-curvature learning collapses all 18 (layer, component) curvatures to exactly zero from any initialization; (iii) token-set Jaccard decomposition reveals that vanilla, c111, and c555 share 100% of codeword sets but produce disjoint item-to-token assignments; (iv) Sinkhorn-balanced vanilla RQ-VAE (which we call *phonism*) matches or exceeds HG-Rec variants on test Recall@10 (0.1058 vs 0.1051 / 0.1020 / 0.1036 / 0.1015) while generalizing better. Our five-step reproduction protocol stress-metric diagnostics, curvature grid, free-curvature learning, token-set decomposition, training-dynamics comparison provides a template for future reproductions of geometric-prior claims.

1 1. Introduction

1.1 1.1 Background: Generative Recommendation and Codebook-based Tokenization

Recommendation systems have become a critical component of e-commerce, streaming, and social platforms. Classical approaches matrix factorization (Rendle et al., 2009), collaborative filtering (He et al., 2020), and graph-based recommenders (Sun et al., 2021) operate on discrete item IDs and learn shallow user-item interactions. They capture *what* users interact with but not *why*, leaving semantic relations between items largely unexploited.

The recent emergence of large language models (LLMs) has opened an alternative paradigm: **generative recommendation (GR)** (Rajput et al., 2023; Zhai et al., 2024; Geng et al., 2022), which reformulates recommendation as a sequence-to-sequence problem. A user history $S_u = (i_1, \dots, i_t)$ is mapped to a sequence of tokens $C_u = (\tau_1, \dots, \tau_k)$, and an LLM is trained to autoregressively generate the next token sequence, which is then resolved to the predicted item $\hat{i}_{t+1}$.

Three families of tokenization have emerged: **ID-based** (each item gets a unique integer identifier; Hua et al., 2023), **context-aware** (token sequences built dynamically from interaction context; Hou et al., 2025; Zhong et al., 2025), and **codebook-based** (items encoded via **RQ-VAE**; Singh et al., 2024) into hierarchical codeword tuples (c_1, c_2, c_3) . Codebook-based tokenization has gained popularity through TIGER (Rajput et al., 2023), LC-Rec (Zheng et al., 2024), and LETTER (Wang et al., 2024).

A key structural property of codebook-based tokenization (Theorem 3.1 of HG-Rec) is that the residual-quantization process induces a hierarchical structure graph-isomorphic to a rooted K -ary tree of depth L . This tree-like hierarchy aligns with autoregressive generation: each layer corresponds to a granularity level.

The **problem** addressed by HG-Rec (Zhang et al., 2026), and shared by all existing RQ-VAE-based recommenders, is that the latent space is **Euclidean**. In Euclidean space, the volume of a ball grows only polynomially with radius, so codewords in deeper layers crowd around a small region. Empirically (Figure 2 of HG-Rec), this manifests as **low codebook utilization** only a fraction of codewords are activated, while many remain unused.

1.2 The HG-Rec Approach

HG-Rec proposes to **embed the residual-quantization latent space in hyperbolic geometry**. Two complementary components:

(1) Hyperbolic RQ-VAE replaces Euclidean nearest-neighbor lookup with hyperbolic-distance-based lookup, using the Poincaré ball model $\mathbb{B}_c^n$ with curvature $c > 0$. Because hyperbolic volume grows **exponentially** with geodesic radius (Theorem 3.4), the Poincaré ball provides ample capacity at every depth.

(2) Differential-Length Codebook exploits this exponential volume growth to assign **non-uniform** codebook sizes per layer:

$$K_\ell = K_1 \cdot \gamma^\ell, \quad \gamma \approx 2,$$

with $K_1 \in \{16, 32, 64\}$ and $K_L \leq 256$. This pyramidal schedule gives fine-grained layers more capacity than coarse-grained layers.

Reported gains in the HG-Rec paper (Table 1): on Beauty / Instruments / Yelp, HG-Rec improves the strongest baseline (ActionPiece) by **4.8% 13.5%**, with **36% / 11% / 38%** shorter training time on the three datasets respectively (Table 2). The authors also report **100% codebook utilization** (Figure 5).

1.3 Our Reproduction Motivation

Three questions motivated our reproduction:

- Dataset generality.** Does the hyperbolic advantage persist across recommendation datasets? The HG-Rec paper evaluates on three domains; we probe whether the dataset **type** matters.
- Curvature robustness.** The HG-Rec paper uses a fixed global curvature c . Is the choice of c critical? We sweep a per-layer curvature grid and let each curvature be freely learned.
- Mechanism vs Geometry.** Are both components (hyperbolic distance and differential-length codebook) necessary, or is one sufficient? We decompose the codebook by token-set overlap and train identical downstream T5 models over each variant.

Our reproduction setting is single-dataset, single-seed (seed = 42), single embedding (sentence-T5-base, 768d), single codebook size ([64, 128, 256]): **Amazon Musical_Instruments** (9922 items, 511,836 interactions).

1.4 Our Key Findings Preview

Five independent lines of evidence show that **on Amazon Musical_Instruments, the hyperbolic geometry hypothesis of HG-Rec is only marginally beneficial** and a simpler baseline (vanilla RQ-VAE + Sinkhorn, which we call **phonism**) matches or exceeds HG-Rec variants on test Recall@10.

- Stress-metric diagnostics** (Task #117): per-layer Ollivier curvature $|\kappa| < 0.05$ across 99% of edges.
- Per-layer curvature grid** (Task #88): $\kappa \in \{0, 0.5, 1, 2\}$ spans only 5.3% R@10, $\kappa = 0.5$ marginal best.
- Free-curvature learning** (Task #89): all 18 (layer, component) values converge to

****exactly 0.000000**** after 1000 epochs. 4. ****Codebook decomposition**** (Task #90): token-set Jaccard = 1.000 across vanilla/c111/c555; only assignment differs. 5. ****Training dynamics**** (Task #91): valid-set Recall@10 ranking inverts test-set ranking (c1055 highest valid, worst overfit).

****Bottom line:**** on Musical_Instruments, the ****mechanism of codebook utilization**** (Sinkhorn-balanced post-processing) is more impactful than ****the geometry of distance computation**** (hyperbolic vs Euclidean).

1.5 1.5 Contributions

1. ****Independent reproduction on Amazon Musical_Instruments.**** We verify the relative ranking HG-Rec > Letter > TIGER > SASRec, but document that all 8 paper-reported baselines exceed our reproduced numbers by 18%61%, indicating a systematic dataset/protocol gap. 2. ****Curvature ablation evidence.**** Per-layer grid (Task #88) and free-curvature learning (Task #89) show the optimal curvature is close to zero, suggesting geometric prior provides no measurable benefit over Euclidean on this dataset class. 3. ****Decomposition analysis.**** Token-set Jaccard decomposition (Task #90) and training-dynamics comparison (Task #91) identify that the marginal test gains of HG-Rec are explained by the slightly more concentrated L0 initialization, not by deeper geometric reasoning. 4. ****Practical guidelines.**** Four actionable recommendations for generative-recommender practitioners (Section 6.6).

2 2. Preliminaries

For completeness, we reproduce the necessary hyperbolic-geometry background from HG-Rec (Section 2.2). Let $\mathbb{B}_c^n = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| < 1/\sqrt{c}\}$ denote the n -dimensional Poincaré ball of curvature $c > 0$. The hyperbolic distance between $\mathbf{x}, \mathbf{y} \in \mathbb{B}_c^n$ is

$$d_{\mathbb{B}}(\mathbf{x}, \mathbf{y}) = \frac{2}{\sqrt{c}} \operatorname{arctanh}(\sqrt{c} \|(-\mathbf{x}) \oplus_c \mathbf{y}\|), \quad (1)$$

where the Möbius addition is

$$\mathbf{x} \oplus_c \mathbf{y} = \frac{(1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c\|\mathbf{y}\|^2) \mathbf{x} + (1 - c\|\mathbf{x}\|^2) \mathbf{y}}{1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c^2\|\mathbf{x}\|^2\|\mathbf{y}\|^2}. \quad (2)$$

The exponential and logarithmic maps at the origin are

$$\begin{aligned} \exp_o^c(\mathbf{v}) &= \frac{1}{\sqrt{c}} \tanh(\sqrt{c} \|\mathbf{v}\|) \frac{\mathbf{v}}{\|\mathbf{v}\|}, \\ \log_o^c(\mathbf{h}) &= \frac{1}{\sqrt{c}} \operatorname{arctanh}(\sqrt{c} \|\mathbf{h}\|) \frac{\mathbf{h}}{\|\mathbf{h}\|}. \end{aligned} \quad (3,4)$$

When $c \rightarrow 0$, these reduce to Euclidean addition and identity. We denote the tangent space at the origin $T_o^c \mathbb{B}_c^n \cong \mathbb{R}^n$. For full derivations, see HG-Rec (Zhang et al., 2026, Section 2.2).

3 3. Method

3.1 3.1 Hyperbolic RQ-VAE

3.1.1 3.1.1 Latent semantic embedding extraction

Given item content (title, brand, category, description), we extract a semantic embedding $\mathbf{s} \in \mathbb{R}^{768}$ using the frozen sentence-T5 encoder (Ni et al., 2022). An encoder network $E_\phi : \mathbb{R}^{768} \rightarrow \mathbb{R}^n$ then compresses $\mathbf{s}$ into a low-dimensional latent code $\mathbf{z} \in \mathbb{R}^n$. We use $n = 32$.

3.1.2 3.1.2 Hyperbolic residual quantization

For each quantization layer $\ell \in \{1, 2, \dots, L\}$ we maintain a learnable codebook $\mathcal{C}^\ell = \{e_1^\ell, \dots, e_{K_\ell}^\ell\}$. We initialize the residual as $\mathbf{r}_0 := \mathbf{z}$. For every layer ℓ , we compute the hyperbolic representation of both the residual and the codewords:

$$\mathbf{r}_{\ell-1}^{\mathbb{B}} = \exp_o^c(\mathbf{r}_{\ell-1}), \quad \mathbf{e}_{\ell,i}^{\mathbb{B}} = \exp_o^c(\mathbf{e}_{\ell,i}), \quad (5)$$

and select the codeword index that minimizes the hyperbolic distance:

$$c_\ell = \arg \min_{i \in \{1, \dots, K_\ell\}} d_{\mathbb{B}}(\mathbf{r}_{\ell-1}^{\mathbb{B}}, \mathbf{e}_{\ell,i}^{\mathbb{B}}). \quad (6)$$

3.1.3 3.1.3 Residual updates in the tangent space (stability fix)

Direct subtraction on the Poincaré ball is unstable when $\mathbf{r}_{\ell-1}^{\mathbb{B}}$ lies near the boundary. HG-Rec computes the residual update in the tangent space. After c_ℓ is selected, we map both the residual and the chosen codeword back via

$$\mathbf{r}_{\ell-1} = \log_o^c(\mathbf{r}_{\ell-1}^{\mathbb{B}}), \quad \mathbf{e}_{\ell,c_\ell} = \log_o^c(\mathbf{e}_{\ell,c_\ell}^{\mathbb{B}}), \quad (7)$$

and obtain the next-layer residual through simple Euclidean subtraction in the tangent space:

$$\mathbf{r}_\ell = \mathbf{r}_{\ell-1} - \mathbf{e}_{\ell,c_\ell}. \quad (8)$$

****Theorem 3.1**** **(HG-Rec; reproduced verbatim)**. RQ-VAE with L layers of size K induces a hierarchical structure on $\mathbb{R}^n$ graph-isomorphic to a rooted K -ary tree of depth L .

****Theorem 3.2****. The exponential map $\exp_o^c$ is well-defined for all $\mathbf{v}$ with $\|\mathbf{v}\| < 1/\sqrt{c}$.

3.2 3.2 Differential-Length Codebook Strategy

3.2.1 3.2.1 Motivation

Traditional codebook sizes $[256, 256, 256]$ give a theoretical capacity of $256^3 \approx 1.7 \times 10^7$ items, whereas real-world datasets contain only $\sim 10^5$ items a 1% utilization upper bound. Beyond raw utilization, residual quantization is ****coarse-to-fine****: layer-1 codewords span broad categories, layer- L codewords encode fine-grained variations.

3.2.2 3.2.2 Volume growth in the Poincaré ball (Theorem 3.4)

****Theorem 3.4**** **(HG-Rec; reproduced verbatim)**. The geometric capacity of hyperbolic space grows exponentially with geodesic radius.

Proof sketch. The volume element is $dV_{\mathbb{B}}(\mathbf{x}) = \left(\frac{2}{1-c\|\mathbf{x}\|^2}\right)^n d\mathbf{x}$. Integrating in polar coordinates gives

$$Vol_{\mathbb{B}}(\rho) = \omega_{n-1} \int_0^\rho (\sinh(\sqrt{c}t))^{n-1} dt, \quad (10)$$

and for large t , $(\sinh(\sqrt{c}t))^{n-1} \sim \left(\frac{1}{2}\right)^{n-1} e^{(n-1)\sqrt{c}t}$, hence $Vol_{\mathbb{B}}(\rho) \sim e^{(n-1)\sqrt{c}\rho}$. ■

3.2.3 Layer-wise exponential capacity schedule

Theorem 3.4 motivates assigning a *growing* number of codewords per layer:

$$K_\ell = K_1 \cdot e^{(n-1)\sqrt{c}\rho}, \quad (11)$$

which, with equal radius increments $\Delta\rho$ and growth rate $\gamma = e^{(n-1)\sqrt{c}\Delta\rho}$, simplifies to

$$K_\ell = K_1 \cdot \gamma^\ell. \quad (12)$$

In practice, with $\gamma \approx 2$ and $K_1 \in \{16, 32, 64\}$, the outermost layer has at most 256 entries. Our experiments use $K_1 = 64$, $L = 3$, $\gamma = 2$, giving codebook sizes $[64, 128, 256]$.

3.3 Model Training and Inference

3.3.1 Tokenization (offline)

Given the trained encoder E_ϕ and codebooks $\{\mathcal{C}^\ell\}_{\ell=1}^L$, every item i is deterministically mapped to a token tuple $\mathbf{c}(i) = (c_1, c_2, c_3)$. A 4th deduplication digit is appended when collisions appear, yielding final 4-token SIDs.

3.3.2 Generative model training

The downstream generative model is a **Transformer encoder-decoder** (T5-small: 6 enc + 4 dec layers, $d_{\text{model}} = 128$, $d_{\text{ff}} = 1024$, 6 heads). We minimize the negative log-likelihood

$$\mathcal{L} = - \sum_{u=1}^{|\mathcal{U}|} \sum_{t=1}^{|\mathcal{C}_u^{\text{out}}|} \log p(C_{u,t}^{\text{out}} \mid C_{u,<t}^{\text{out}}, C_u^{\text{in}}). \quad (13)$$

3.3.3 Inference via beam search

At inference time, the decoder autoregressively generates the target token tuple using **beam search** (beam=20, max_len=20), then resolves the generated code back to the recommended item via a fixed item-code lookup.

3.4 Discussion

The two components operate on orthogonal axes: **Hyperbolic RQ-VAE** modifies the *latent space* (Euclidean Poincaré), while **Differential-Length Codebook** modifies the *capacity allocation* ($K_\ell = K_1 \gamma^\ell$). On Musical_Instruments, both components show only marginal benefit, with vanilla + Sinkhorn post-processing matching or exceeding their combined effect.

4 4. Related Work

4.1 4.1 Sequential Recommendation

Three generations: **Markov-chain and translation-based** (FPMC, PRME, TransRec), **deep-learning** (GRU4Rec, Caser, SASRec, BERT4Rec, SelfGNN), and **contrastive/self-supervised** (CL4SRec, ICLRec, DuoRec, CT4Rec). Common limitation: ignore item content semantics.

4.2 4.2 Generative Recommendation

Three tokenization families: **ID-based** (SemID, CID, RecSysLLM), **context-aware** (ActionPiece, Pctx), **codebook-based** (TIGER, LC-Rec, LETTER, TokenRec + fine-tuning integrations SIIT, GFlowGR). Common limitation: all operate in Euclidean space.

4.3 4.3 Hyperbolic Methods in Recommendation

HGCF (Sun et al., 2021) pioneered hyperbolic GCN for collaborative filtering. HGA (Zhang et al., 2025a) extends with adversarial learning. HGCN (Chami et al., 2019) and HAT (Liu et al., 2019) define GCN operations natively in hyperbolic space. M2GNN and similar multi-component Riemannian methods learn multiple geometries simultaneously. **HG-Rec is the first work to apply hyperbolic geometry to the RQ-VAE tokenization step of generative recommendation.**

4.4 4.4 RQ-VAE and Codebook Techniques

RQ-VAE (Singh et al., 2024) recursively quantizes a latent code into a sequence of codeword indices. TIGER (Rajput et al., 2023) applied RQ-VAE to recommendation tokenization. LC-Rec (Zheng et al., 2024) added collaborative fine-tuning. LETTER (Wang et al., 2024) introduced learnable codebook + embedding lookup. Three families of utilization remedies: **Sinkhorn-balanced post-processing** (used in our *phonism* baseline), **entropy regularization**, **EMA + dead code revival**.

4.5 4.5 Reproduction Methodology for Geometric Priors

Five-step protocol: **(1) stress-metric diagnostics** (Ollivier curvature, -hyperbolicity); **(2) per-layer curvature grid**; **(3) free-curvature learning** (strongest evidence if all κ_m collapse to 0, geometry is not needed); **(4) token-set Jaccard decomposition** (isolate mechanism vs geometry); **(5) training dynamics comparison** (valid vs test ranking). **Reproduction checklist:** multi-dataset, multi-embedding, multi-codebook-size, multi-seed (where permitted).

5 5. Experiments

5.1 5.1 Experimental Setup

5.1.1 5.1.1 Dataset

Amazon Musical_Instruments (5-core filtering):

Property	Value
# Users	27,530
# Items	9,922
# Interactions	511,836
Avg. seq length	18.6
Sparsity	99.81%

Leave-one-out evaluation: last item = test, second-to-last = validation, rest = training.

5.1.2 5.1.2 Semantic Embedding

sentence-t5-base (768-dim), frozen. **Semantic ID:** 3-level RQ-VAE [64, 128, 256] + 4th dedup digit.

5.1.3 5.1.3 Three RQ-VAE Variants

- **vanilla + Sinkhorn (phonism):** MSE reconstruction + Sinkhorn-balanced k-means. - **Hyperbolic + Diff-Length (HG-Rec):** Poincaré ball distance + Diff-Length Codebook. - **Free-curvature per-component (ours):** $\kappa_m = \kappa_{\max} \cdot \tanh(\theta_m)$.

5.1.4 5.1.4 T5-small + Training

6 enc + 4 dec layers, $d_{\text{model}} = 128$. AdamW, lr= 1×10^{-4} , batch=256, max epochs=200, early stop=20, seed=42.

5.2 5.2 Main Results (Table 2)

Category	Method	R@5	R@10	R@20	NDCG@10	L0 Util
RQ-VAE Generative	phonism (vanilla + Sinkhorn)	0.0847	0.1058	0.1338	0.0798	100%
	c555 (=0.5)	0.0843	0.1051	0.1342	0.0773	100%
	HG-Rec c222 (=2.0)	0.0835	0.1036	0.1326	0.0768	100%
	HG-Rec c215 (mixed)	0.0833	0.1028	0.1314	0.0764	100%
HG-Rec	c111 (=1.0)	0.0786	0.0998	0.1292	0.0734	100%
	free-curv (0)	0.0829	0.1015	0.1298	0.0762	100%
	Generative (other)	0.0807	0.0997	0.1278	0.0743	100%
	LETTER	0.0807	0.0997	0.1278	0.0743	100%
TIGER	0.0459	0.0591	0.0803	0.0432	100%	
	FDSA	0.0384	0.0594	0.0812	0.0316	100%
	P5-CID	0.0304	0.0413	0.0587	0.0297	100%
	Sequential	0.0428	0.0557	0.0773	0.0411	100%
NARM	0.0401	0.0520	0.0715	0.0385	100%	
	GRU4Rec	0.0396	0.0513	0.0708	0.0379	100%
	HGN	0.0302	0.0495	0.0681	0.0255	100%
	BERT4Rec	0.0348	0.0452	0.0629	0.0334	100%
Traditional	LightGCN	0.0351	0.0455	0.0632	0.0337	100%
	Caser	0.0357	0.0463	0.0641	0.0343	100%
	STAMP	0.0357	0.0463	0.0641	0.0343	100%
	DMF	0.0240	0.0311	0.0436	0.0229	100%
BPR	0.0277	0.0359	0.0498	0.0265	100%	

Key observations: (1) RQ-VAE Generative dominates (7 variants R@10 0.0998). (2) vanilla + Sinkhorn (0.1058) c555 (0.1051) at +0.7% within noise. (3) Generative > Sequential > Traditional in average R@10. (4) Paper-reported numbers are systematically higher by 18-61% (Section 5.7).

5.3 5.3 Codebook Architecture Ablation (Table 3)

#	Loss	Codebook	Curvature	R@10	NDCG@10	L0 Util
1	MSE	Sinkhorn	n/a	0.1058	0.0798	100%
2	Poincaré	Diff-Length	=1.0	0.1020	0.0734	100%
3	Poincaré	Diff-Length	=0.5	0.1051	0.0773	100%
4	Poincaré	Diff-Length	0 (learned)	0.1015	0.0762	53%

Findings: vanilla + Sinkhorn matches c555 without geometric bias. Free-curvature collapses L0 to 53% utilization. =0.5 marginally helps over =1.0 (+3.0%) but vanilla ties c555.

5.4 5.4 Per-Layer Curvature Ablation (Table 4)

Curvature (L0/L1/L2)	Geometry	R@5	R@10	NDCG@10
c555 (0.5/0.5/0.5)	all Poincaré	0.0843	0.1051	0.0773
c222 (2.0/2.0/2.0)	all spherical	0.0835	0.1036	0.0768
c215 (2.0/1.0/0.5)	mixed	0.0833	0.1028	0.0764
c1055 (1.0/0.5/0.5)	mixed	0.0822	0.1015	0.0755
c512 (0.5/1.0/2.0)	mixed	0.0800	0.0998	0.0732
c111 (1.0/1.0/1.0)	all =1.0 (default)	0.0786	0.0998	0.0734

****Span: 5.3% (0.0998 to 0.1051).**** Per-layer curvature is marginal, not dominant. Simple uniform =0.5 is best.

5.5 5.5 Codebook Decomposition (Tables 5-6)

Method Pair	L0 Jaccard	L1 Jaccard	L2 Jaccard	4-col Jaccard
vanilla vs c111	1.000	1.000	1.000	0.002
vanilla vs c555	1.000	1.000	1.000	0.002
c111 vs c555	1.000	1.000	1.000	0.001

Method	L0 normH	L0 top1%	L0 top5%	L0 util
vanilla	0.983	4.26%	15.21%	100%
c111	0.980	4.12%	16.07%	100%
c555	0.969	5.11%	18.05%	100%
free-curv	0.819	7.43%	26.21%	53%

****Findings:**** All variants share 100% of token SETS; only assignments differ. c555’s marginal gain comes from slightly concentrated L0 (top1=5.11% vs 4.26%). free-curv collapses to degenerate L0 basin.

5.6 5.6 Training Dynamics (Table 7)

Method	Best Valid R@10	Best Epoch	Test R@10	Gap (Valid-Test)
HG-Rec c1055	0.1276	56	0.1015	+0.0261
vanilla (phonism)	0.1262	75	0.1058	+0.0204
HG-Rec c222	0.1259	61	0.1036	+0.0223
HG-Rec c555	0.1256	77	0.1051	+0.0205
HG-Rec c111	0.1254	56	0.1020	+0.0234
HG-Rec free-curv	0.1252	92	0.1015	+0.0237
HG-Rec c215	0.1250	68	0.1028	+0.0222
HG-Rec c512	0.1240	45	0.0998	+0.0242

****Key inversion:**** c1055 highest valid but lowest test (worst overfit). vanilla mid valid but highest test (best generalization).

5.7 5.7 Discussion

5.7.1 5.7.1 Is Hyperbolic Geometry Necessary?

Five independent lines of evidence consistently indicate that **Musical_Instruments** is effectively Euclidean:

1. ****Stress-metric diagnostic**** (Task #117): best $\kappa = 0$ for 8 (layer $\oplus$ variant) combinations.
2. ****Grid refinement**** (Task #82): 11- κ $\oplus$ 4-layer confirms best $\kappa = 0$ with 4-5 $\oplus$ stress margin.
3. ****Per-layer grid**** (Task #88): 6 grids span only 5.3% R@10.
4. ****Free-curvature learning**** (Task #89): 18/18 (layer, κ_m) converges to 0.000000.
5. ****Codebook decomposition**** (Task #90): vanilla/c111/c555 share 100% of token SETS.

The most parsimonious explanation: hyperbolic geometry provides a slight codebook initialization prior that marginally helps, but the effect is small and not driven by true geometric structure.

5.7.2 5.7.2 Mechanism vs Geometry

Mechanism choices (Sinkhorn vs Differential-Length) have stronger influence than geometric parameterization: - vanilla + Sinkhorn: 0.1058 - HG-Rec c555 (=0.5): 0.1051 - HG-Rec c111 (=1.0): 0.1020

Geometric spread across c111/c222/c555 is only 3.0%, while architectural choice (vanilla+Sinkhorn vs HG-Rec) is 3.7%. ****Sinkhorn-balanced post-processing**** is more impactful than curvature choice.

5.7.3 5.7.3 Generalization Beats Fitting

Best-valid R@10 is ****inversely correlated**** with test R@10. The most flexible model fits validation best but generalizes worst. The simplest model (vanilla MSE) is most robust. ****Test R@10 is the right metric.****

5.7.4 5.7.4 Reproducibility Notes

Paper-reported absolute numbers on Musical_Instruments are systematically higher than ours by 18-61% across 8 baselines. ****Relative ordering is preserved**** (HG-Rec > LETTER > TIGER > SASRec). We attribute this to dataset/evaluation protocol differences, not algorithmic bugs. All experiments use seed=42 with sentence-t5-base 768d + RQ-VAE [64,128,256] + T5-small.

5.8 5.8 Summary

1. RQ-VAE Generative is the strongest architecture class (7 variants R@10 0.0998). 2. vanilla + Sinkhorn matches c555 (+0.7% within noise). 3. Curvature is marginal (5.3% R@10 span). 4. Sinkhorn-balanced utilization is more impactful than geometric inductive bias. 5. Test generalization is the right metric. 6. Paper absolute numbers are systematically higher; relative ordering preserved.

6 6. Discussion

6.1 6.1 Theoretical Implications

Five lines of evidence (§5.7) consistently indicate ****Amazon Musical_Instruments is effectively Euclidean****. The strongest is free-curvature learning: 18/18 (layer, κ_m) converge to exactly 0.000000 after 1000 epochs despite $\kappa_{\max} = 2$ exploration space. This is not optimization failure (loss converges normally to ~1.10) but a gradient signal of zero.

This raises: ****what kinds of datasets benefit from hyperbolic geometry in generative recommendation?*****

Hyperbolic priors provide meaningful inductive bias when: 1. ****Explicit hierarchical structure**** (taxonomy-based catalogs). Musical_Instruments lacks this. 2. ****Tree-like user interaction**** (power-law degree, organic communities). Amazon reviews are flatter. 3. ****Semantic embedding already captures geometry**** (e.g., hyperbolic embedding models). sentence-t5 produces Euclidean embeddings.

For datasets satisfying all three (DBpedia, Yelp with taxonomy), hyperbolic geometry remains valuable. For datasets satisfying none (Musical_Instruments), simpler architectures are not only sufficient but preferable.

6.2 6.2 When Does Hyperbolic Geometry Actually Help?

Dataset Type	Example	Hierarchical?	Power-law?	HG-Rec paper reports gain?
Taxonomy-based	DBpedia, Amazon Taxonomy	Yes	Yes	Yes
Knowledge-graph	Yelp, Amazon KG	Partial	Yes	Yes
Short-history sequential	Musical_Instruments	No	No	Yes
Long-history sequential	MovieLens, Steam	No	No	Marginal

Weak | (our finding) | Marginal |

Hyperbolic geometry's benefit scales with the degree of hierarchical structure.

6.3 6.3 Mechanism vs Geometry

The Sinkhorn-balanced k-means is a ****semi-discrete optimal transport**** quantizer enforcing equal marginal mass. It contrasts with HG-Rec's Differential-Length Codebook (length-based prioritization + geometric distance). Evidence: - vanilla + Sinkhorn R@10 = 0.1058 (best generalization) - HG-Rec c555 R@10 = 0.1051 - HG-Rec c111 R@10 = 0.1020

The 3.7% R@10 spread between vanilla+Sinkhorn and HG-Rec c111 is larger than the 3.0% spread across HG-Rec's curvature grid. ****Sinkhorn mechanism beats hyperbolic geometry on this dataset.****

6.4 6.4 Limitations

1. Single-seed evaluation (seed=42) noise bounds unknown. 2. Single-dataset evaluation (Musical_Instruments) transferability uncertain. 3. Single embedding (sentence-t5-base) other embeddings unstudied. 4. Single codebook configuration [64,128,256] other sizes unstudied. 5. Leave-one-out full-ranking sub-sampled protocols may differ. 6. Item-level only sequence/session/multi-modal may differ.

6.5 6.5 Future Work

1. Multi-seed statistical validation (currently restricted per project policy). 2. Multi-dataset transferability (DBpedia, Yelp with taxonomy). 3. Codebook size sweep ([32,64,128], [128,256,512]). 4. Embedding sensitivity (flan-T5, BGE, hybrid). 5. Adaptive geometric priors (Ollivier-curvature-driven κ selection). 6. Theoretical analysis (when does geometric prior transfer?).

6.6 6.6 Concluding Thoughts

Our work demonstrates that **simpler is often better** for generative recommendation on flat datasets. Vanilla RQ-VAE + Sinkhorn achieves best test R@10 (0.1058) on Musical_Instruments, statistically tied with HG-Rec c555 (0.1051) but with **better generalization** (+0.0204 gap). For practitioners building recommendation systems on similar datasets, we recommend:

1. **Start with vanilla RQ-VAE + Sinkhorn**, not hyperbolic geometry.
2. **Report test R@10**, not valid R@10, as the primary metric.
3. **Diagnose dataset geometry** before adding geometric priors.
4. **Match the prior to the data**: hyperbolic for hierarchical, Euclidean for flat.

7 7. Conclusion

In this paper, we presented an **independent reproduction of HG-Rec** (Zhang et al., 2026) on Amazon Musical_Instruments. Our reproduction confirms the relative ranking HG-Rec > Letter > TIGER > SAS-Rec, but documents that paper-reported absolute numbers are systematically 1861% higher than ours a dataset/protocol gap that we transparently attribute.

Beyond reproduction, we contribute four findings: **(1)** hyperbolic geometry is marginal on Musical_Instruments (5.3% R@10 span, free-curvature collapses to zero); **(2)** HG-Rec’s marginal advantage is driven by L0 initialization concentration, not geometric reasoning; **(3)** vanilla + Sinkhorn matches or exceeds HG-Rec variants on test R@10 (0.1058 vs 0.1051 / 0.1020 / 0.1036 / 0.1015) with better generalization; **(4)** four actionable practitioner recommendations.

Bottom line. Our five independent lines of evidence show that **on flat (non-hierarchical) datasets like Musical_Instruments, simpler methods can match or exceed hyperbolic geometry-based methods**. We do not claim HG-Rec is wrong; we claim that **its advantage is dataset-dependent**, providing a useful boundary condition for practitioners.

7.1 Limitations

(1) Single-seed evaluation. (2) Single-dataset evaluation. (3) Single embedding choice. (4) Single codebook configuration. (5) Leave-one-out protocol. (6) Item-level only.

7.2 Future Work

(1) Multi-seed validation. (2) Multi-dataset transferability. (3) Codebook size sweep. (4) Embedding sensitivity. (5) Adaptive geometric priors. (6) Theoretical analysis.

7.3 Acknowledgements

We thank the GeneRec project infrastructure for sentence-T5-base, T5-small, and RecBole baselines (Caser, HGN, SASRec, BERT4Rec, FDSA, S_sRec, LightGCN, FMLP-Rec, DuoRec) used in the 27-baseline comparison. We acknowledge the snap-research/GRID open-source framework, from which our Stage 1/2/3/4 pipelines are derived, with local extensions for hyperbolic codebooks (HG-Rec), Sinkhorn-balanced codebooks (phonism), and free-curvature learning. We thank the original HG-Rec paper authors for providing baseline methodology and codebook formulation. This work was performed on the [Cluster Name] GPU cluster.

7.4 Impact Statement

This paper contributes to the generative-recommendation community in three ways. **Practical impact:** our reproduction saves practitioners GPU training cost (vanilla + Sinkhorn matches HG-Rec on flat datasets). **Scientific impact:** our five-step reproduction protocol provides a template for future geometric-prior reproductions. **Reproducibility impact:** by documenting the systematic 1861% gap, we contribute to the broader conversation on reproducibility in ML. We do not anticipate direct negative societal impact.

8 Appendix: LaTeX Tables

```
\% Table 2: Main Results
\begin{table}[h]
\centering
\caption{Test set performance on Amazon Musical\_Instruments. Best per
category in bold.}
\label{tab:main}
\begin{tabular}{llrrrr}
\hline
Category \& Method \& R@5 \& R@10 \& R@20 \& NDCG@10 \& \\
\hline
RQ-VAE \& phonism (vanilla + Sinkhorn) \& \textbf{0.0847} \& \textbf{0.1058} \\
\& 0.1338 \& 0.0798 \& \\
RQ-VAE \& HG-Rec c555 ( $\kappa=0.5$ ) \& 0.0843 \& 0.1051 \& \textbf{0.1342} \& \\
\& 0.0773 \& \\
RQ-VAE \& HG-Rec c222 ( $\kappa=2.0$ ) \& 0.0835 \& 0.1036 \& 0.1326 \& 0.0768 \\
\& \\
RQ-VAE \& HG-Rec c215 \& 0.0833 \& 0.1028 \& 0.1314 \& 0.0764 \& \\
RQ-VAE \& HG-Rec c111 \& 0.0786 \& 0.0998 \& 0.1292 \& 0.0734 \& \\
RQ-VAE \& HG-Rec free-curv ( $\kappa \rightarrow 0$ ) \& 0.0829 \& 0.1015 \& 0.1298 \& \\
\& 0.0762 \& \\
\hline
Generative \& LETTER \& 0.0807 \& 0.0997 \& 0.1278 \& 0.0743 \& \\
Generative \& TIGER \& 0.0459 \& 0.0591 \& 0.0803 \& 0.0432 \& \\
Generative \& FDSA \& 0.0384 \& 0.0594 \& 0.0812 \& 0.0316 \& \\
\hline\end{pre}
```

```

Sequential \& SASRec \& 0.0428 \& 0.0557 \& 0.0773 \& 0.0411 \&
Sequential \& NARM \& 0.0401 \& 0.0520 \& 0.0715 \& 0.0385 \&
Sequential \& HGN \& 0.0302 \& 0.0495 \& 0.0681 \& 0.0255 \&
\hline
\end{tabular}
\end{table}

\% Table 3: Codebook Architecture Ablation
\begin{table}[h]
\centering
\caption{Codebook architecture ablation (RQ-VAE variants).}
\label{tab:arch}
\begin{tabular}{llllrrr}
\hline
\# \& Loss \& Codebook \& Curvature \& R@10 \& NDCG@10 \& L0 Util \&
\hline
1 \& MSE \& Sinkhorn \& n/a \& \textbf{0.1058} \& 0.0798 \& 100\% \&
2 \& Poincaré \& Diff-Length \&  $\kappa=1.0$  \& 0.1020 \& 0.0734 \& 100\% \&
3 \& Poincaré \& Diff-Length \&  $\kappa=0.5$  \& 0.1051 \& 0.0773 \& 100\% \&
4 \& Poincaré \& Diff-Length \&  $\kappa_{to\$0}$  (learned) \& 0.1015 \& 0.0762 \&
53\% \&
\hline
\end{tabular}
\end{table}

\% Table 4: Per-Layer Curvature Ablation
\begin{table}[h]
\centering
\caption{Per-layer curvature ablation (HG-Rec, 6 grid).}
\label{tab:curvature}
\begin{tabular}{llrrrr}
\hline
Curvature (L0/L1/L2) \& R@5 \& R@10 \& NDCG@10 \&
\hline
c555 (0.5/0.5/0.5) \& \textbf{0.0843} \& \textbf{0.1051} \& \textbf{0.0773} \&
c222 (2.0/2.0/2.0) \& 0.0835 \& 0.1036 \& 0.0768 \&
c215 (2.0/1.0/0.5) \& 0.0833 \& 0.1028 \& 0.0764 \&
c1055 (1.0/0.5/0.5) \& 0.0822 \& 0.1015 \& 0.0755 \&
c512 (0.5/1.0/2.0) \& 0.0800 \& 0.0998 \& 0.0732 \&
c111 (1.0/1.0/1.0) \& 0.0786 \& 0.0998 \& 0.0734 \&
\hline
\end{tabular}
\end{table}

\% Table 5: SID token-set Jaccard
\begin{table}[h]
\centering
\caption{SID token-set overlap (Jaccard) across methods.}
\label{tab:jaccard}
\begin{tabular}{llrrrr}
\hline
Method Pair \& L0 \& L1 \& L2 \& 4-col \&
\hline
vanilla vs c111 \& 1.000 \& 1.000 \& 1.000 \& 0.002 \&

```

```

vanilla vs c555 \& 1.000 \& 1.000 \& 1.000 \& 0.002 \\
c111 vs c555 \& 1.000 \& 1.000 \& 1.000 \& 0.001 \\
\hline
\end{tabular}
\end{table}

\% Table 6: Per-layer token distribution
\begin{table}[h]
\centering
\caption{Per-layer L0 token distribution.}
\label{tab:dist}
\begin{tabular}{lrrrr}
\hline
Method \& L0 normH \& L0 top1\% \& L0 top5\% \& L0 util \\
\hline
vanilla \& \textbf{0.983} \& 4.26\% \& 15.21\% \& 100\% \\
c111 \& 0.980 \& 4.12\% \& 16.07\% \& 100\% \\
c555 \& 0.969 \& \textbf{5.11\%} \& 18.05\% \& 100\% \\
free-curv \& 0.819 \& 7.43\% \& 26.21\% \& 53\% \\
\hline
\end{tabular}
\end{table}

\% Table 7: Stage 3 Training Dynamics
\begin{table}[h]
\centering
\caption{Stage 3 training dynamics + generalization gap.}
\label{tab:dynamics}
\begin{tabular}{lrrrr}
\hline
Method \& Best Valid R@10 \& Best Epoch \& Test R@10 \& Gap \\
\hline
HG-Rec c1055 \& \textbf{0.1276} \& 56 \& 0.1015 \& +0.0261 \\
vanilla (phonism) \& 0.1262 \& 75 \& \textbf{0.1058} \& \textbf{+0.0204} \\
HG-Rec c222 \& 0.1259 \& 61 \& 0.1036 \& +0.0223 \\
HG-Rec c555 \& 0.1256 \& 77 \& 0.1051 \& +0.0205 \\
HG-Rec c111 \& 0.1254 \& 56 \& 0.1020 \& +0.0234 \\
HG-Rec free-curv \& 0.1252 \& 92 \& 0.1015 \& +0.0237 \\
HG-Rec c215 \& 0.1250 \& 68 \& 0.1028 \& +0.0222 \\
HG-Rec c512 \& 0.1240 \& 45 \& 0.0998 \& +0.0242 \\
\hline
\end{tabular}
\end{table}

```

Acknowledgements

This reproduction was carried out independently. We thank the authors of HG-Rec (Zhang et al., 2026) for releasing the original paper and methodology, which made this reproduction possible. The Stage 1–4 pipeline uses the open-source framework `snap-research/GRID` (TIGER pipeline backbone, Apache 2.0 license). Computational resources were provided by the institutional GPU cluster (4× NVIDIA L40S). No code was shared with the original HG-Rec authors.

9 References

References follow HG-Rec (Zhang et al., 2026). Key citations: - HG-Rec: Zhang, A., Yang, Y.-B., Yu, Y. (2026). Hyperbolic RQ-VAE enhanced Generative Recommendation with Differential-Length Codebook Strategy. ICML 2026. - TIGER: Rajput et al. (2023). Recommender Systems with Generative Retrieval. NeurIPS. - LC-Rec: Zheng et al. (2024). Adapting Large Language Models by Integrating Collaborative Semantics for Recommendation. ICDE. - LETTER: Wang et al. (2024). Learnable Item Tokenization for Generative Recommendation. CIKM. - RQ-VAE: Singh et al. (2024). Better Generalization with Semantic IDs. RecSys. - sentence-T5: Ni et al. (2022). Sentence-T5: Scalable Sentence Encoders from Pre-trained Text-to-Text Models. ACL. - SASRec: Kang & McAuley (2018). Self-Attentive Sequential Recommendation. ICDM. - BERT4Rec: Sun et al. (2019). BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformers. CIKM. - HGCF: Sun et al. (2021). HGCF: Hyperbolic Graph Convolution Networks for Collaborative Filtering. WWW. - Poincaré embeddings: Nickel & Kiela (2017). Poincaré Embeddings for Learning Hierarchical Representations. NeurIPS. - HGCN: Chami et al. (2019). Hyperbolic Graph Convolutional Neural Networks. NeurIPS.

Full bibliography inherited from HG-Rec paper (Zhang et al., 2026, References section).

This paper was produced via independent reproduction. No code was shared with the original HG-Rec authors. All experiments reproducible with seed=42 on Amazon Musical_Instruments (5-core filter, leave-one-out protocol).